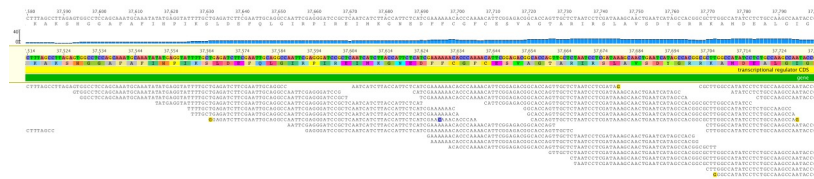


Some solutions to challenges in metagenomic assembly

Mapping metagenomic reads to a reference genome



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Some solutions to challenges in metagenomic assembly

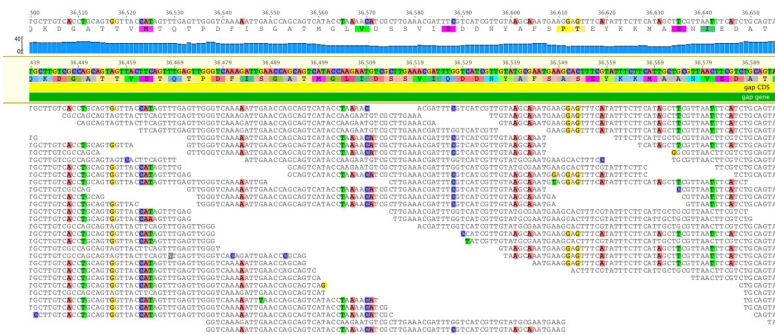
Mapping metagenomic reads to a reference genome



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Some solutions to challenges in metagenomic assembly

Mapping metagenomic reads to a reference genome



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Some solutions to challenges in metagenomic assembly

- Binning by coverage and GC content
- Multisized derbujin graph

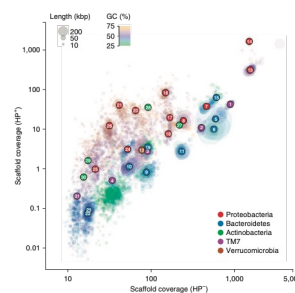
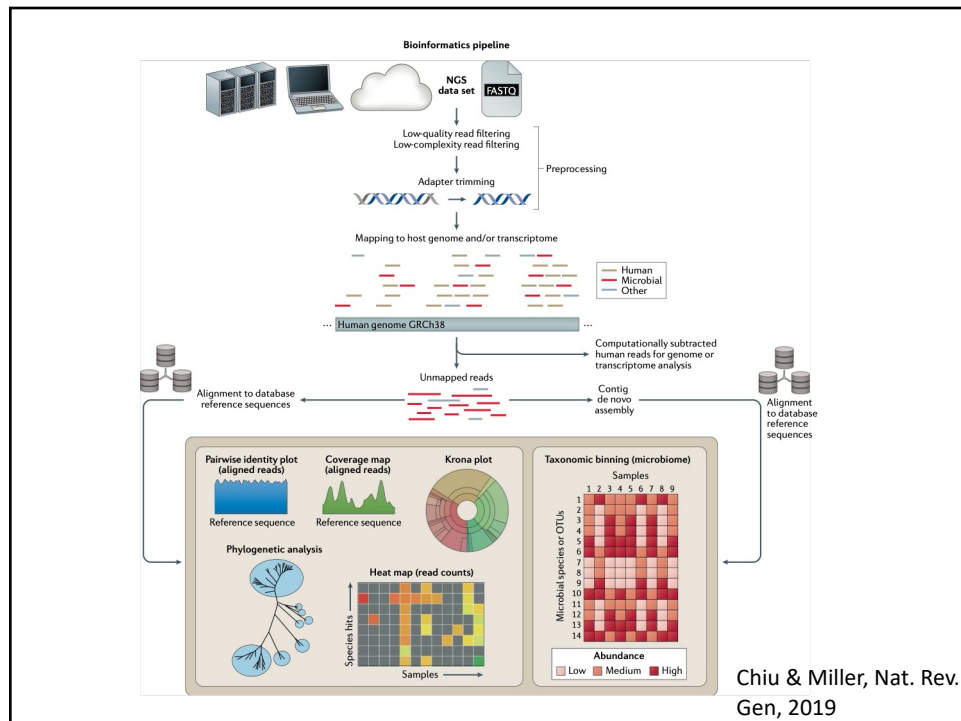


Figure 1 Sequence composition-independent binning of metagenome scaffolds from the lab-scale bioreactor using differential coverage (HP², HP²). Circles represent scaffolds, scaled by the square root of their length and colored by GC content. Only scaffolds 25 kbp are shown. Clusters of similarly colored circles represent potential genome bins, the centroids of which are indicated by numbered circles and colored according to phylum-level taxonomic affiliation (Table 1 and Supplementary Table 2). This differential coverage plot provides the starting point for secondary refinement and finishing of genome assemblies (Fig. 2).

Genome sequences of rare, uncultured bacteria obtained by differential coverage binning of multiple metagenomes

Mads Albertsen¹, Philip Hugenholtz^{2,3}, Adam Skarshewski², Kåre I. Nielsen¹, Gene W. Tyson^{2,4} & Per H. Nielsen¹

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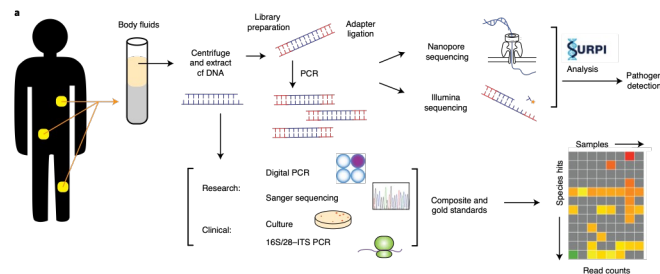


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It works!

Rapid pathogen detection by metagenomic next-generation sequencing of infected body fluids

Wei Gu^{1,2,10}, Xianding Deng^{1,2,10}, Marco Lee^{3,10}, Yasemin D. Sucu^{1,2}, Shaun Arevalo^{1,2}, Doug Stryke^{1,2}, Scot Federman^{1,2}, Allan Gopez^{1,2}, Kevin Reyes^{1,2}, Kelsey Zorn⁴, Hannah Sample⁴, Guixia Yu^{1,2}, Gurpreet Ishpuniani^{1,2}, Benjamin Briggs^{1,2}, Eric D. Chow⁴, Amy Berger⁵, Michael R. Wilson^{6,7}, Candace Wang^{1,2}, Elaine Hsu¹, Steve Miller^{1,2}, Joseph L. DeRisi^{4,8} and Charles Y. Chiu^{1,2,9}



Gu et al, Nat. Med. 2021

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Table 2 | Case series of body fluid mNGS testing in patients with probable or established infection but negative clinical microbiological testing

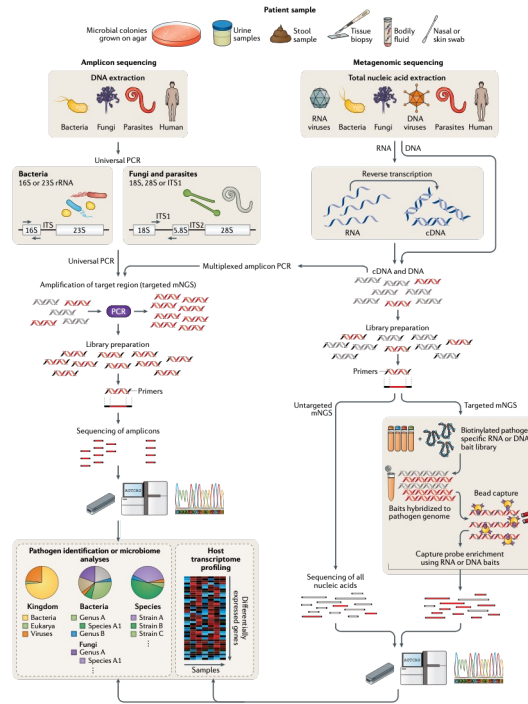
Case no.	Body fluid sample type	Presentation	Body fluid mNGS result	Body fluid culture result	Body fluid 16S/28S-ITS PCR result ^a	Clinical microbiological diagnosis
S88	CSF	Encephalopathy without known cause; has a brain implant	Positive (<i>K. aerogenes</i>)	Negative	Negative	<i>K. aerogenes</i> : same organism grown in culture from surgically removed DBS; concordant dPCR; matches clinical context
S89	Retroureine fluid	Abdominal fluid collection and elevated WBC; a history of abdominal surgery	Positive (multiple gastrointestinal anaerobes) ^b	Negative	ND	None: mNGS results consistent with clinical context
S90	Pleural	Fever, cough, bacteremia, loculated pleural effusion	Positive (<i>S. pneumoniae</i>)	Negative	ND	<i>S. pneumoniae</i> : concordant positive blood culture matches the clinical context
S91	Pleural	Fever, bacteremia, pneumonia, pleural effusion	Positive (<i>S. pyogenes</i>)	Negative	ND	<i>S. pyogenes</i> : concordant positive blood culture matches the clinical context
S92	BAL	Pulmonary nodules post chemotherapy	Positive (<i>A. fumigatus</i>)	Negative	ND	<i>A. fumigatus</i> : probable invasive aspergillosis matches clinical context; serum beta-D-glucan and serum galactomannan positive
S176	BAL	Cavitary lesion of the lung	Positive: below threshold (<i>C. immitis</i>)	Negative	ND	<i>C. immitis</i> : serum coccidioides antibody positive, 1:16 titer on complement fixation
S177	CSF	Chronic meningitis	Positive (<i>Cladophiala bantiana</i>)	Negative	Negative	<i>C. bantiana</i> : brain tissue culture is positive for rare <i>C. bantiana</i>
S178	Pleural fluid	Cavitary lung lesions and pleural effusion	Negative ^c	Negative	ND	<i>C. neoformans</i> : BAL and bronchial wash fluid culture positive for <i>C. neoformans</i> ; serum cryptococcal antigen positive
S179	CSF	Headache, vision changes and optic disc edema	Positive: <i>T. pallidum</i> (below threshold)	ND	Negative	<i>T. pallidum</i> : serum rapid plasma reagin, venereal disease research laboratory and treponemal antibody positive
S180	Pleural fluid	Lymphadenopathy, lymphocytic pleural effusion	Negative ^c	Negative	ND	MTB: positive purified protein derivative; lymph node biopsy; necrotizing granulomas, AFB stain positive and MTB PCR positive
S181	CSF	Headaches, blurred vision, night sweats, neck stiffness	<i>Candida parapsilosis</i>	Negative	Negative	<i>C. parapsilosis</i> : serum beta-D-glucan >500 pg ml ⁻¹ ; CSF culture 24 d later positive for <i>C. parapsilosis</i>
S182	CSF	Headache, photophobia, blurred vision, hydrocephalus	Negative ^c	Negative	Negative	<i>Sporothrix schenckii</i> : CSF (>1:16) and serum (1:32) <i>Sporothrix</i> antibody positive

^aBacterial 16S PCR or fungal 28S-ITS PCR. ^bThe top five anaerobes were *Forciobacterium prausnitzii*, *Eubacterium rectale*, *Akkermansia muciniphila*, *Acidimicrobium intestini* and *Bifidobacterium adolescentis*. ^cInfectious diagnosis missed by mNGS testing. ND, not done.

Gu et al, Nat. Med. 2021

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Clinical Metagenomics (Methods)



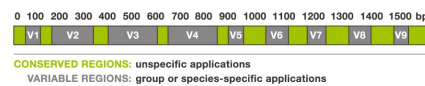
Chiu & Miller, Nat. Rev. Gen., 2019

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New methods are needed for Microbiome Analysis

Uniting the classification of cultured and uncultured bacteria and archaea using 16S rRNA gene sequences

Pablo Yarza^{1,2,3}, Pelin Yilmaz², Elmar Pruesse², Frank Oliver Glöckner^{2,4}, Wolfgang Ludwig², Karl-Heinz Schleifer⁵, William B. Whitman⁶, Jean Euzéby⁷, Rudolf Amann⁸ and Ramon Rosselló-Móra¹



www.alimetrics.net

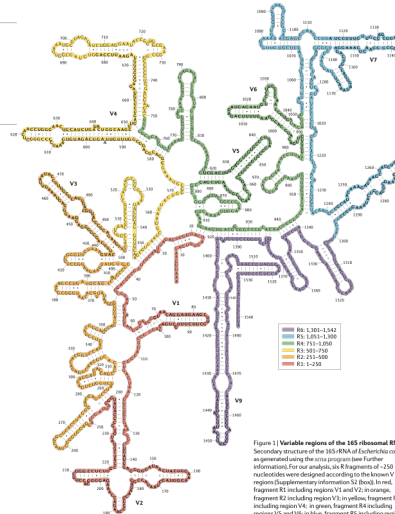


Figure 1 Variable regions of the 16S ribosomal RNA. Secondary structure of the 16S ribosomal RNA gene, as generated using the rna program from the Rfam database. For our analysis, the 16S ribosomal RNA gene was fragmented into 150 nucleotides were designed according to the known V regions (Supplementary information S1, Table S1, and Table S2). Fragment R1 including regions V1 and V2, in orange; fragment R2 including regions V3 to V9, in purple; fragment R3 including regions V1, V3, V5, V6, V7, V8, and V9, in blue; fragment R4 including regions V1 and V2, in green; fragment R5 including regions V1 and V2, in red; fragment R6 including regions V1 and V2, in pink; fragment R7 including regions V1 and V2, in brown; fragment R8 including regions V1 and V2, in grey.

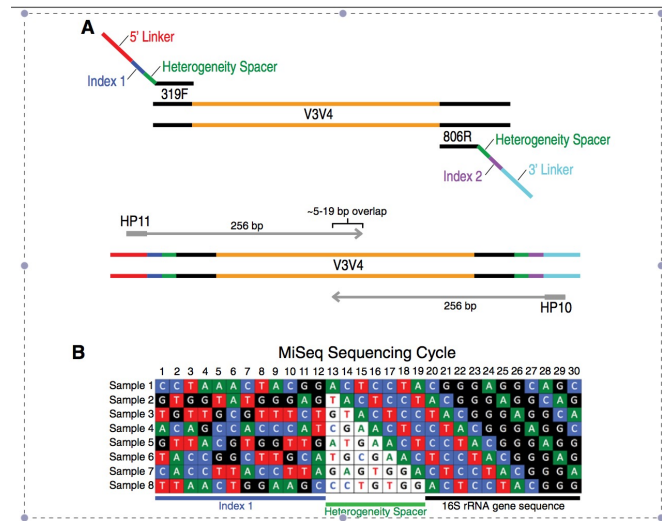
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Challenges in Amplicon Sequencing

- How to deal with errors?
- Multiplexing and de-multiplexing 100s of samples with very similar sequences
- Assigning sequences to taxa
- Comparison of samples

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Some Solutions (multiplexing)



Fadrosh, et al, Microbiome, 2014

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Some Solutions (errors)

```

ATGCTCGCAGGCTCCTCTTCA
ATGCTCGCAGGCTCCTCTTCA
ATGCTCGCAGGCTCCTCTTCA
ATGCTCGCAGGCTCCTCTTCA
ATGCTCGCAGGCTCCTCTTCA
ATGCTCGCAGGCTCCTCATCA
ATGCTCGCAGGCTCCTCTTCA
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ATGCTCGCAGGCTCCTCTTCA
ATGCTCGCAGGCTCCTCTTCA

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Operational Taxonomic Unit (OTU)

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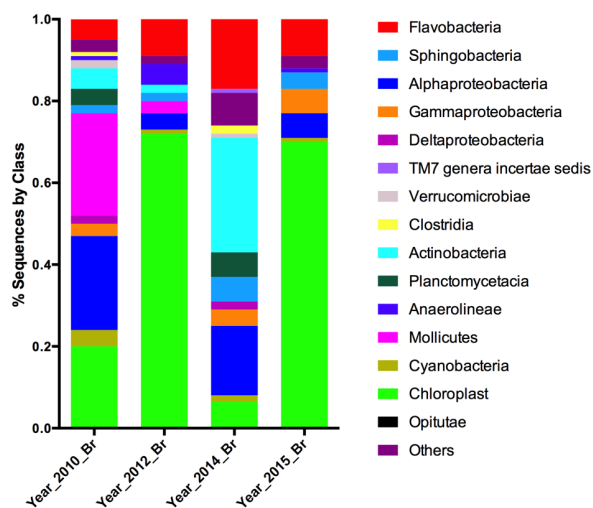
Some Solutions (assignment)

Creation of 16S rDNA reference sequence databases (Green Genes, RDP, Silva) to facilitate assignment of taxa



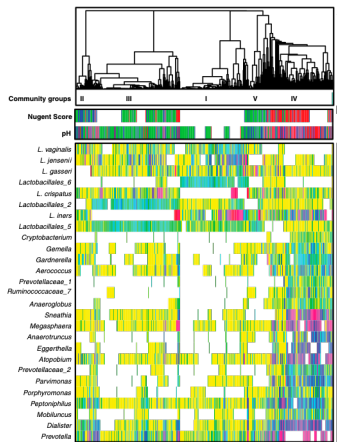
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Output



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Some Solutions (large scale comparison)



Vaginal microbiome of reproductive-age women

Jacques Ravel^{1,†}, Pawel Gajer², Zaid Abdo³, G. Maria Schneider⁴, Sara S. K. Koenig⁵, Stacey L. McCulle⁶, Shara Karlebach⁷, Reshma Gorle⁸, Jennifer Russell⁹, Carol O. Tacket¹, Rebecca M. Brotman¹⁰, Catherine C. Davis¹¹, Kevin Ault¹², Ligia Peralta¹³, and Larry J. Forney¹⁴

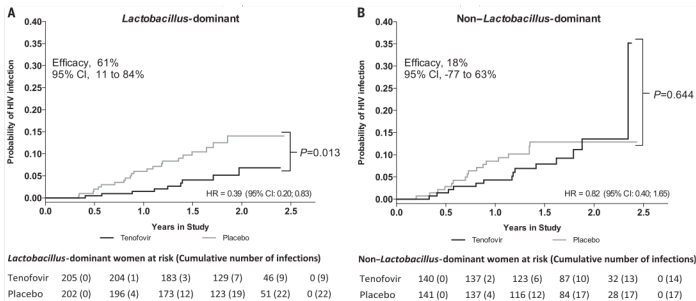
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It matters!

HIV CLINICAL TRIALS

Vaginal bacteria modify HIV tenofovir microbicide efficacy in African women

Nichole R. Klatt,^{1,†} Ryan Chu,^{1,†} Kenzie Birse,^{2,3,†} Alexander S. Zevin,^{1,†} Michelle Perner,^{2,3,†} Laura Noè-Romas,^{2,3,†} Anneke Grobler,⁴ Garrett Westmacott,⁵ Irene Y. Xie,^{2,3} Jennifer Butler,^{2,3} Leila Mansoor,⁶ Lyle R. McKinnon,^{2,4} Jo-Ann S. Passmore,^{4,6} Quarraisha Abdool Karim,^{1,7} Salim S. Abdool Karim,^{4,7} Adam D. Burgener^{2,3,8,†}



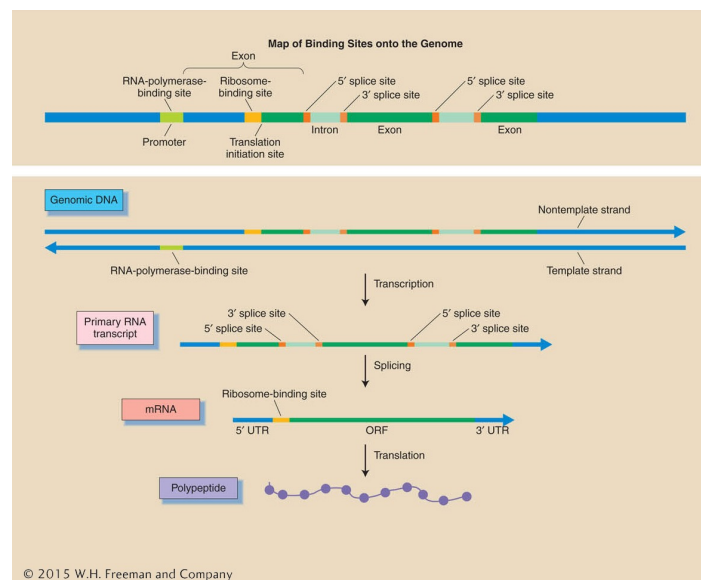
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RNA Sequencing (RNA-Seq)

- Used to obtain gene expression information (not all genes are expressed all the time!)
 - Which genes are expressed in which conditions?
 - Which genes are expressed in which cells?
 - Which genes are expressed in which tissues?
 - Detects isoforms

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Genome Annotation



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